Project Charter: The Impact of Weak Authentication on IoT Devices Security

Background

As more and more devices are connected to the Internet of Things (IoT), they have become extremely vulnerable to cybercrime attacks, mostly due to weak or default authentication methods. The project aims to provide a structured approach to reducing vulnerabilities in IoT networks caused by weak authentication, identifying the most critical vulnerabilities and implementing methods to mitigate these risks.

Goals

1. To develop a clear research question and clearly articulate the extent of the research investigation of IoT authentication vulnerabilities at the end of the preparation phase.
2. Carry out a preliminary investigation, one using structure of primary, and one using structure of secondary reports, to establish a solid groundwork.
3. Develop an empirical research method to collect relevant and usable cybersecurity information, identify a set of specific survey questions and add a brief outline of a simulation project to generate information.

Scope

This specific project charter culminates in the completion of all the basic preparatory and planning frameworks needed to inform technical aspects of the project to follow.

In-Scope

Work 1 (preparation): Identifying the main research topic and wording the exact parameter of the research scope formally.

Work 2 (Planning): Conduct preliminary research on finding out basic information for both the Primary and Secondary Report, creating and designing the Research

Methodology/Simulation project, draft the empirical survey questions, set timelines and milestones, Organize the Gantt chart, and finalize this Project Charter.

Out-of-Scope

Work 3 (Development): This charter excludes from the scope of development activities conducted by the participants such as in-depth literature review, threat modeling, vulnerability assessment, case study analysis, data analysis, threat calculation, security solution development and proof of concept development.

Work 4 (Testing): Technical tasks such as threat/tuning analysis, sensitivity, locating specific solutions, getting expert reviews, performing academic reviews and management of recommendations not included.

Work 5 (Release): released work, in the form of comprehensive research paper written, presentation created, published work monitored and disseminated strategy run.

Key Stakeholders

There are two types of stakeholders. They are internal stakeholder and external stakeholder.

For this project, internal stakeholders are project manager, project assistant, data analysis and department coordinator.

For this project, external stakeholders are project client, project sponsor, cybersecurity consultant and cybersecurity faculty advisor.

Project Milestones

- Project start date is May 15, 2026.
- Work 1(Preparation) is complete as the focus and scope of the research are fixed (in late May 2026).
- June 23, 2026 will be the project charter sign-off date for Planning Phase (Work 2).
- Invoicing Dates to Client: Invoicing of final project deliverables for basic planning will be done on client sign-off, June 23, 2026.

Project Budget

For Non-Recurring Expenses:

Expense Category	Item Description	Cost
Hardware Infrastructure	IoT Testbed Devices: Raspberry Pi 4 microcomputers, wireless routers, smart home appliances and firmware flashing adapters	£260.40
Software License	On top of that, there are Core Network Vulnerability Scanners like OpenVAS by GreenBone, Nmap Scripting Engine and OWASP ZAP.	£0.00
Specialized Technical Software	The dedicated software utilities, as well as specific plug-in extensions and API tokens	£111.60

Expense Category	Item Description	Cost
	from companies (used in credential stuffing attacks and dictionary attack databases)	
Total Project Budget		£372.00

Hardware Cost from sale sites:

Raspberry Pi 4 microcomputers [https://www.raspberrypi.com/products/raspberry-pi-4-model-b] (18.6.2026)

Wireless Router [https://ict.com.mm/collections/wireless-routers] (18.6.2026)

Firmware Flashing Adapters [https://haloidsolutions.com/products/vertex-standard-fif-8a-fif8-flash-rom-adapter-firmware-writer?srsltid=AfmBOooZ2LpmouxMsOCIdBdubxy72sTHWbhPkHaCbZxcF2lJlU515AtB]

Software License:

Greenbone [https://www.greenbone.net/en/products]

Nmap [https://nmap.org/]

For Monthly Recurring Expenses:

Expense Category	Item Description	Monthly Cost (GBP / £)
Cloud Sandbox Compute	Amazon Web Services (AWS) EC2 instances to safely stage and simulate malware remotely on standard, 2 vCPUs and 4 GB RAM general-purpose environments.	£23.00 per month
Hosted Vulnerability Scanning	A base level subscription to a pro level cloud security scanning suite that allows for tracking of up to 5 target IP/ Domains (with example of HostedScan Basic tier or other lightweight SaaS-based tracking suite).	£36.00 per month
Subtotal Recurring Costs		£59.00 per month

Monthly Recurring Expenses:

AWS [https://aws.amazon.com/pricing/?nc2=h_pr_hub]

Constraints

Time Resource Constraint: All the preparation/panning activities should be completed within the agreed schedule, which is due on June 23,2026.

Phased Focus Constraint: Only WBS Work 1 and Work 2 are the Project works in which the project must be active under this charter, technical development and live testing will not be allowed.

Assumptions

Project Assistant Resources: The Project Assistant will be completely booked to conduct primary/secondary report data collection and development of survey questions on time.

Academic Alignment: The proposed research methodology and the Cyber security simulation design are assumed to be offered timely by the Cyber securities Faculty Advisor.

Risks and Dependencies

HR Risks: If members of the team (including the Project Assistant) have conflicting timelines or become delayed, the question and methodology layouts could be delayed from being delivered.

Time Resource Risk: Execution window is tight and the time sensitivity could not be greater as the window of time for the execution lies between May 15 to June 23.

Budget and Tool Dependency: The software and hardware required and the money for them are critical to advancing to the next phase of implementation or any subsequent phase.

Approval Signatures

		Mrs.Thiri Win Win
[Name], Project Client	[Name], Project Sponsor	Mrs.Thiri Win, Project Manager